

Group 9 error analysis.

Introduction

The model which had the best accuracy was one of our RoBERTa models, with an accuracy of 0.755. This is the model we used to get the predictions for our error analysis. We got the sentences, the “True Emotions” (which were our labels), the “Predicted Emotions” (predictions from the model), and the original sentences from each instance where the “True Emotion” and “Predicted Emotion” were not the same. These were our misclassifications. These were put into a table and each misclassification was put into one or more categories depending on what we believed caused the misclassification. We originally had 5 classifications, but throughout the process added two more.

Classifications

These 7 classifications are:

1 Contextual Misunderstandings – When seeing words without any context, they could have a myriad of meanings. Same with sentences, they could mean multiple things. If it was obvious that the sentence had a different meaning in the context of these transcriptions from how it is most commonly used, the misclassification would be classified here.

2 Inter-Class Similarity – This classification is for misclassifications which occurred due to the sentence in question being able to mean different common things, and as such being able to exert different emotions, but there is no or very little context which points to its use in this emotion.

3 Intra-Class Variability – This is a classification for when the sentence is very uncommonly used for this emotion and the context does not support its use in this emotion.

4 Occlusions – These were classified when certain singular words present in the sentence seemingly cause the model to put the emotion differently from what it was.

5 Truth Misclassification – There is a question whether our “True Emotions” were the ground truth or not, and since they were also classified by an AI, mistakes could, and did, happen. If the “True Emotions” was wrong and “Predicted Emotions” was correct, the misclassification would be classified here.

6 Translation Error – Since we translated our sentences from Polish to English, translation errors are bound to occur. Alongside that, nuances were lost and the same

words in Polish and English could have more meanings in one of the two languages (polysemy), causing the meaning to change.

7 Meta – These were sentences which were not actual sentences. Any instance of “MUSIC”, “APPLAUSE”, or a censored sentence, if present and misclassified, would be classified here.

Results

From out 196 misclassifications, we counted 206 mistakes (since one misclassification could be caused by multiple causes simultaneously), split up as follows:

Context	Inter	Intra	Occlusion	Truth	Translation	Meta
74	33	7	29	31	15	20
35,4%	15,8%	3,4%	13,9%	14,8%	7,2%	9,6%

Since “Truth” and “Meta” are not errors of our model, but instead failures earlier in the pipeline, we will remove them from this analysis, to entirely focus this effort on the misclassifications of our model. This leaves us with 155 mistakes, split up as follows:

Context	Inter	Intra	Occlusion	Translation
74	33	7	29	15
47,7%	21,3%	4,5%	18,7%	9,7%

The overwhelming number of mistakes were “Contextual Misunderstandings”. These were mostly short sentences (“Yes.”, “Oregano.”, “Why?”, “Okay.”) which on their own could be several different emotions. In context, it is relatively clear which emotion these sentences should be classified as. Of course, emotions are subjective and as such, such short sentences are subject to harder interpret the emotion of, even for humans, so it is understandable that the model may struggle with this.

The next largest number of mistakes is “Inter-Class Similarity”. These are sentences which could go either the way of the “True Emotion” or “Predicted Emotion”, and the context leaves it ambiguous.

With a little bit less mistakes, “Occlusions” are the 3rd biggest cause of misclassifications here. The label “Occlusions” itself has a bit of a different meaning than how it was used here, as it was used to classify sentences in which singular words pushed the model to classify it differently than what the entire sentence meant.

After that is the classification “Translation Error”. This is for sentences where the translation creates sentences which either don’t make sense in English or lose their nuance or meaning, leading to the model to misclassify them. A simple example of such a sentence losing nuance is “Synu.” in Polish, which translates to “Son.” in English. In

Polish, son can be and is used more informally than in English. You could theoretically remove this classification as well, as we did with “Truth Misclassification” and “Meta”, as there is an argument that this too is a failure earlier in the pipeline, but since some mistakes could have been rectified by the model, it was decided to keep it in.

The final and smallest classification is “Intra-Class Variability”, where sentences are used in a way that is rarely used with the specific emotion, and the context doesn’t show this.

Conclusion

From our analysis, the biggest cause of mistakes by our model are short sentences in which the context does not seem to be used to classify the emotion. Two other big causes are ambiguous sentences, which even with context aren’t a clear emotion, and singular words in a sentence which are commonly used in a different emotion than the sentence is implying.